u_ism_and_star_birth

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1 ISM and Star Formation

In this packet, we turn to the study of stellar formation. The details of stellar formation and evolution depend sensitively on masses and compositions. To get oriented, we start with the formation of stars that are very similar to our Sun, after which, we can turn to stars that are much bigger, smaller, older, past their primes, *et cetera*.

Throughout this packet, we will assume that the stars we are talking about have masses and metalicities similar to the Sun.

In AA:SS, we studied the formation of the Sun from a fragment of a Giant Molecular Cloud (GMC). Rather than redo the whole thing for very similar stars, we will review some aspects of that process only very briefly and focus on other aspects that we did not emphasize in `ss_solar_system_1,2,3`. For students who did not take AA:SS and would like to know more, I recommend you look over those packets and/or take AA:SS next year. For students who did take AA:SS, I recommend you review your notes from that class. Of course, I am always happy to answer any questions you may have about * homologous cloud collapse and fragmentation * core heating throughout stages * accretion mechanisms * anything else

1.1 Interstellar Matter

In space we notice the stars, and to a lesser extent the planets, but there is *much* more material there that we don't see. We call this *interstellar matter* or ISM. In fact, we do "see" ISM when we know how to look. ISM is roughly divided into three categories:

- atomic (size ~ 1 AA = 10^{-10} m = 0.1 nm)
- molecular (size ~ 10 AA = 10^{-9} m = 1 nm)
- dust (size ~ 1000 AA = 10^{-7} m = 0.1 μm)

The dust does not emit light, but it reflects/absorbs light making for dark spots in the sky where otherwise there would be light. The interactions of particles with light are collectively called **scattering** and the processes through which this happens are called **scattering** processes. The diminution of light by dust is called **extinction** and **reddening**. In the first effect, light is re-reflected away from the line of sight and lost into space. Particles with sizes d much larger than the wavelength of the light λ simply block the light and cast a shadow with an area denoted by σ called the **cross section** of the particle.

1.1.1 Rayleigh scattering

is the is the scattering of light by particles with size $d \ll \lambda$ is much smaller than the wavelength of the light. The details are complicated but the formula for the intensity of the scattered light is

$$I = cI_0 \left(\frac{1 + \cos^2 \theta}{2} \right) \frac{d^6}{r^2} \frac{1}{\lambda^4} \qquad (\lambda \gg d) \tag{1}$$

This is called the Rayleigh formula for scattering. Here c is some dimensionless constant, I_0 is the intensity of the light before it is scattered, r is the distance to the particle doing the scattering (note the characteristic r^{-2} fall-off!), and θ denotes the direction in which the light is scattered. This formula says that light is scattered in all directions and says what fraction goes in each direction.

1.1.2 Problem 1

a. Explain why the radial dependence for the intensity falls off like $1/r^2$. (Hint: this is the fall-off of intensity with distance we've discussed before.)

b. Plot the $\frac{1}{2}\left(1 + \cos^2 \theta\right)$ factor in Desmos to see what the radiation field looks like around the scattering particle.

c. Note the strong dependence on the wavelength and use it to explain why the sun is white/yellow, the sky is blue, and sunsets are red. Calculate a range for how much more strongly blue light is scattered compared to red.

d. For visible light, to what type of interstellar matter does the Rayleigh solution apply?

1.1.3 Answer 1

a. This is the usual $1/r^2$ fall-off of any massless force carrier. We saw this behavior for both the gravitational force law and the Coulomb force.

b. The angular distribution of Rayleigh-scattered radiation. The curve has been exaggerated to $(1 + a \cos^2 \theta)$ with $a = 4$ to emphasize the double-lobe structure.

c. The characteristic λ^{-4} dependence implies that short-wavelength light is affected far more strongly than long-wavelength light. For example, the visible spectrum runs from 380 `nm` (violet) to 750 `nm` (red), so violet is scattered $\left(\frac{750}{380}\right)^4 (1.97)^4 = 15$ times more than red.

If we looked directly at the Sun (don't do it!!) it would look white, but if we look off to some random point in the clear sky, it would look blue, because this is the color that has the best chance of being scattered back to us. During sunsets, on the other hand, we generally look somewhere near the direction of the Sun. Now we are looking through a lot of atmosphere so the Sun is no longer white, but red, since the blue light was mostly scattered out on it's way to our eyes.

Below, we compare the scattering of blue and red light quantitiatively.

d. For visible light, $\lambda \sim 500\ nm = 5 \times 10^{-7}\ m \gg d$, so d must be the size of atoms or molecules. Dust grains are roughly 5 times smaller than this, so a different approximation would be more appropriate for them, however, the applicable Mie theory is far more complicated than we should get into here.

[1]:
```
import astropy.units as u
```

```
print(f"\n|   ANSWER 1\n|")

blue_low = 450*u.nm
blue_high = 485*u.nm
red_low = 635*u.nm
red_high = 750*u.nm

print(f"|   1c. For blue light between {blue_low}-{blue_high} and red between␣
↪{red_low}-{red_high},\n|       the scattering is anywhere between {(red_low/
↪blue_high)**4:.0f} and {(red_high/blue_low)**4:.0f} times as strong.\n")
```

```
|   ANSWER 1
|
|   1c. For blue light between 450.0 nm-485.0 nm and red between 635.0 nm-750.0
nm,
|       the scattering is anywhere between 3 and 8 times as strong.
```

Along these lines, shorter-wavelength light is scattered more and longer wavelength light is scattered less. Thus, the light that has passed through an interstellar cloud has had more blue light removed (reradiated in all directions) and the remainder is reddish, just like it is for sunsets on Earth.

1.1.4 Problem 2

Meet Barnard 68. It is a cloud of dust and molecules 125 pcs away in the constellation Ophiuchus. Explain why there are no stars in this hole and why those around the edges of the hole are red.

1.1.5 Answer 2

A thick dust cloud is blocking (scattering) the visible light. The tell-tale sign of this is the reddening around the edge where the dust cloud thins and some of the longer-wavelength visible light makes it through. The shorter-wavelength visible light is still scattered away, making the color red.

As we discussed in class, there are three main types of interstellar cloud depending on what the hydrogen is doing. Hydrogen comes in three forms:

- H I: Neutral hydrogen atom consisting of a proton nucleus and an electron bound to it in one of infinitely many orbitals indicated by the "principal quantum number" $n = 1, 2, 3, \ldots$. These orbitals have definite energy given by the formula

  $$E_n = -13.6\ \mathtt{eV}\frac{1}{n^2} \qquad n \in \mathbf{N} \tag{2}$$

  The $-$ sign indicates that the electron is bound to the proton. (If the energy of an electron is positive, it is unbound and will escape the proton.) We see that as we increase $n \to \infty$, the electron is getting closer and closer to freedom as $E_n \to 0$. Thus, there are infinitely many bound orbitals but any energy in excess of 13.6 eV is enough to kick an electron in any orbital out of the atom. When that happens, we get

- H II: Ionized hydrogen. This is a hydrogen ion that is missing its electron, that is, a prosaic term for the word "proton".

Going the other way, a hydrogen atom can bind to another hydrogen atoms to make

- H2 : Molecular hydrogen. This is two hydrogen atoms bound together covalently by the valence electrons of each HI. The covalent binding energy is about 4.5 eV.

An aside on notation: Spectroscopists refer to various states of ionization of an element, let's call it "Em" with a roman numeral: Em I is the neutral version of the element Em, Em II is the singly-ionized one (so it has charge +1), Em III is doubly-ionized, and so on. These ionization states can be identified by the changes in their spectral lines.

WARNING: These conventions are *not* the same as those you learned in chemistry! Make sure you know the difference, because this is *astrophysics* class.

1.1.6 Problem 3

a. Fe IX and Fe X are ions present in the sun's atmosphere. What are the charges of these ions?

b. What's wrong with the ions H III and He IV?

1.1.7 Answer 3

a. X is the Roman symbol for 10 and IX makes it one fewer, so the charges are +8 for FeIX and +9 for FeX.

b. H only has one electron, so HIII would have -1 electrons, which is impossible. Similarly, HeIV would have 3 electrons removed, but it only has at most 2 in the first place.

The major discoveries of the Quantum Mechanics revolution included the realization that particles and waves are two sides of the same coin: Yes an electron is a particle, but it also has wave-like properties like (de Broglie) wavelength. Yes, light is a wave, but it comes in discrete packets called "photons", so it is also a particle with a definite energy given by the Einstein-Planck relation

$$E = hf \qquad \text{with} \qquad h = 6.6 \times 10^{-34}\ J \cdot s$$

A particle of light has a very specific speed $c = 3.0 \times 10^8\ m/s$ when it travels in vacuum. This gives a relation for the wavelength in terms of the frequency

$$\lambda = \frac{c}{f} \tag{4}$$

1.1.8 Problem 4

Use these relations to calculate the wavelength of light

a. needed to dissociate the hydrogen molecule into two hydrogen atoms

b. needed to ionize a hydrogen atom

c. least energetic photon that results from a transition to the ground state $n = 1$

d. the Hα line. This is the first Balmer spectral line, which means that is the least energetic transition to the first excited state $n = 2$.

e. Identify which part of the electromagnetic spectrum these photons correspond to (*i.e.* visible, UV, IR, *etc.*)

(Recall that one electronvolt is 1.6×10^{-19} J.)

[2]:
```python
from astropy.constants import c, h
from aa_tools.astroconst import spectrum_wavelength

print(f"\n|   ANSWER 4\n|")

def wavelength (dE: u.Quantity) -> u.Quantity:
    _lambda = h*c/dE
    return _lambda.decompose()

dE_covalent = 4.5 * u.eV
dE_H = 13.6 * u.eV
dE_12 = dE_H*(1-1/2**2)
dE_23 = dE_H*(1/2**2-1/3**2)

print(f"|   a. Recall that the molecular hydrogen covalent bond has an energy␣
↪of\n|      {dE_covalent} = {dE_covalent.to(u.J):.1e}. This corresponds to a␣
↪wavelength of {wavelength(dE_covalent).to(u.nm):.0f},\n|      which is␣
↪{spectrum_wavelength(wavelength(dE_covalent))}. \n|")

print(f"|   b. The ionization energy of the hydrogen atom is {dE_H} = {dE_H.to(u.
↪J):.1e}.\n|      This corresponds to a wavelength of {wavelength(dE_H).to(u.
↪nm):.0f}, which is {spectrum_wavelength(wavelength(dE_H))}. \n|")

print(f"|   c. This is the Lyman-alpha line with energy {dE_12} = {dE_12.to(u.J):
↪.1e}.\n|      This corresponds to a wavelength of {wavelength(dE_12).to(u.nm):.
↪0f}, which is {spectrum_wavelength(wavelength(dE_12))}. \n|")

print(f"|   d. The hydrogen-alpha transition has energy {dE_23:.1f} = {dE_23.
↪to(u.J):.1e}.\n|      This corresponds to a wavelength of {wavelength(dE_23).
↪to(u.nm):.0f}, which is {spectrum_wavelength(wavelength(dE_23))}. \n|")

print(f"|   e.  See above.\n")
```

```
|   ANSWER 4
|
|   a. Recall that the molecular hydrogen covalent bond has an energy of
|      4.5 eV = 7.2e-19 J. This corresponds to a wavelength of 276 nm,
|      which is ultraviolet.
|
|   b. The ionization energy of the hydrogen atom is 13.6 eV = 2.2e-18 J.
|      This corresponds to a wavelength of 91 nm, which is ultraviolet.
|
|   c. This is the Lyman-alpha line with energy 10.2 eV = 1.6e-18 J.
|      This corresponds to a wavelength of 122 nm, which is ultraviolet.
|
|   d. The hydrogen-alpha transition has energy 1.9 eV = 3.0e-19 J.
```

```
|      This corresponds to a wavelength of 656 nm, which is visible (red).
|
|   e.  See above.
```

Note that this calculation implies that it doesn't take much to break up molecules.

The characterization of interstellar clouds can now be further refined. Firstly, consider a dense molecular cloud. It contains dust grains, molecules, and atoms all mixed together. Generally, hot young stars are forming in these regions, and these stars emit copious amounts of high-energy radiation. The higher energies do not penetrate into the darkest parts of the cloud. We start deep in such a cloud and work our way outward toward a star from there.

We start in regions with dust and molecular hydrogen (and other molecules). Often the molecular hydrogen H2 will bind through van der Waals forces to the dust, and the dust will shield it from dissociation by the high-energy radiation. These regions can be quite cool at around 10 `K`. The dominant cooling mechanism is **molecular line emission**: the ability of molecules to emit photons by changing their vibrational and rotational states. Unfortunately, H2 is essentially invisible, because it is so small and symmetrical: It is made of two identical atoms so the center of mass is at the geometric center and there is not asymmetry possible in its charge distribution, so it has no electric dipole moment. This is important because to change from one rotational state the next lower one would require a photon to carry away one unit of angular momentum. However, photons can only do so by coupling to the dipole moment, which H2 does not have.

What we need to look for is a different, less symmetrical molecule and try to use that instead. Such a molecule is called a tracer, and they are generally present in the ISM whenever conditions are favorable for H2 formation. The most prevalent tracer for the dense, cool clouds we are discussing here is CO. When it transitions from it's angular momentum $J = 1$ state to the non-rotating $J = 0$ ground state, a 2.6 `mm` photon is emitted.

1.1.9 Problem 5

a. Assuming this transition sets the scale for the temperature of the cloud, convert this photon energy to a temperature to confirm that it gives the correct order of magnitude.

Continuing on our journey out of the molecular cloud toward a star, we next encounter the **photodissociation region** (PDR): the region where the penetrating far-UV radiation has sufficiently high energy to dissociate H2 into H I atoms (problem 4a). The temperature in this region is determined by the ... to be in the range 50 - 1000 `K` range. With the higher end closer to the star and the lower end matching up with the dense molecular cloud, many layers are in the 100s of `K` range. While the far-UV radiation tends to heat the gas, molecular line emission from C II and O I cool it. The wavelengths of these modes are in the 10s-100s **microns** ($1\ \mu m = 10^{-6}\ m$).

1.1.10 Problem 5 (cont'd)

b. Use this to confirm that the claimed temperature of the PDR the correct order of magnitude.

Finally, we get to regions in which the H I is exposed to the high-energy radiation which ionizes it, or **photoionization** region. This is the region where the dominant form of hydrogen is H II. This region is characterized by a balance between the ejected electron kinetic energy heating the gas, and the cooling by various ionic transitions: The naïve conversion of the 91 `nm` wavelength of the

least energetic **photoionizing** photon to a temperature is misleading for two reasons. Firstly, the kinetic energy of the ejected electron is $hf + E_1 = hf - 13.6$ eV and not 13.6 eV. Secondly, there is again a different cooling mechanism, this time predominantly from a forbidden O III transition line.

1.1.11 Problem 5 (cont'd)

c. As mentioned in class, this line is green. Use this fact to estimate an upper limit for the temperature of the PDR.

[3]:
```
from astropy.constants import k_B

print(f"\n|   ANSWER 5\n|")

def temperature(l: u.Quantity) -> u.Quantity:
    _t = h*c/(k_B*l)
    return _t

l_CO = 2.6*u.mm
l_micron = 100e-6*u.m
l_green = 530*u.nm


print(f"|   a. A wavelength of {l_CO} corresponds to a temperature of␣
↪{temperature(l_CO).to(u.K):.1f}.\n|      This is indeed in the 10 K range.\n|")

print(f"|   b. A wavelength of 100 micron = {l_micron:.1e} corresponds to a␣
↪temperature of {temperature(l_micron).to(u.K):.0f}.\n|      This is indeed in␣
↪the 100 K range.\n|")

print(f"|   c. Green is the 500-565 nm range. Picking a wavelength of 530 nm␣
↪gives a\n|      temperature of {temperature(l_green).to(u.K):.1e}.\n")
```

```
|   ANSWER 5
|
|   a. A wavelength of 2.6 mm corresponds to a temperature of 5.5 K.
|      This is indeed in the 10 K range.
|
|   b. A wavelength of 100 micron = 1.0e-04 m corresponds to a temperature of
144 K.
|      This is indeed in the 100 K range.
|
|   c. Green is the 500-565 nm range. Picking a wavelength of 530 nm gives a
|      temperature of 2.7e+04 K.
```

The last estimate is a bit high, but of the right dex. The typical H II region temperatures usually quoted are in the range 7,000-10,000 K.

Above, we motivated the various regions by starting deep in a dense GMC and then progressing toward a star so that successively more energetic light results in various states of dissociation and ionization. However, the atomic and molecular clouds described could also be considered as astrophysical clouds in their own right. Below, we recapitulate our considerations in problem 5, but as separate astrophysical phenomena.

- **Emission nebulae** are clouds of ionized gas that are radiating the light absorbed from stars in the nebula. These are generally red due to the Hα line. There are two main types which are completely different:
  - **Stellar nurseries**: H II regions that are ionized from the UV radiation of new stars
  - **Planetary nebulae**: Ionized gas remnants of dead/dying stars.[^1]
- **Reflection nebulae**: Clouds of ISM that reflect light from nearby stars. Since short wavelength light is scattered more than long wavelength light, these nebulae tend to be bluish in color.
- **Atomic/molecular clouds** come in two quite different types:
  - Molecular clouds' main hydrogen content is in the form of molecular hydrogen H2. They are usually dilute and cool ~ 10 K.
  - Atomic clouds are also called H I regions. Their temperatures can be in a wide range centered around an average of hundreds of Kelvin.
- **Absorption nebulae** or **dark nebulae** are dust clouds that contain complex molecules like dust grains and are positioned in front of an optical source. Such clouds often look like holes in the sky because they scatter all light in the visible range away from our line of sight, although they tend to have reddish edges. Infrared and radio wave light may still pass through.

1.1.12 Dipole interactions and the 21 cm line

We now come back to an important point, about H I regions that we skipped over before. A hydrogen atom in its ground state consists of a proton in the nucleus and an electron in the lowest orbital. As mentioned in class, there are actually two states with closely-related energies in this configuration: The spin of the electron and proton could be aligned or anti-aligned. In `x_spectroscopy`, we described a spinning charged particle as a tiny magnet with a magnetic dipole moment given by the Bohr magneton

$$\mu_B = \frac{g}{2}\frac{q}{m}s\hbar \tag{5}$$

Here q is the charge of the particle, m is its mass, and s is its (half-integer) spin. The gyromagnetic ratio g is equal to $g = 1$ "classicaly", but for a fundamental spin-1/2 ($s = 1/2$) particle like the electron, $g = 2$. Although this has not mattered to us before, the proton-although it too has $s = 1/2$-is *not* a fundamental partcile in the same sense as the electron. Instead, it is a composite of very strongly-interacting quarks and gluons in very complex configurations. In fact the exact structure of the proton is an active area of hadronic physics/strongly-coupled Quantum Chromodynamics (QCD). Experimentally, the gyromagnetic moment of the proton is known to high accuracy: $g_p = 5.5856946893(16)$ whereas theoretically, lattice QCD and other techniques can only match this to the few % level.

As we discussed in `x_spectroscopy`, the interaction energy of a magnetic dipole $\vec{\mu}$ and a magnetic field $\vec{B}$ is given by $U = \vec{\mu}\cdot\vec{B} = \mu B\cos(\theta)$. (This ranges from $+\mu B$ (aligned) to $-\mu B$ (anti-aligned), so the energy gap has a size $\Delta E = 2\mu B$.) However, we have also mentioned that a dipole acts like

a little magnet, so it sets up its own magnetic field (in fact a *dipole field* by definition).

1.1.13 Problem 6

In this multi-part problem, we will explore dipoles and their interactions, with the goal of understanding the 21 cm line. Start by opening this simple on-line demo for electric charges and fields.

6a. We will make an electric dipole field by adding to monopole fields (Coulomb charges) and letting the separation between the charges become small.

i. Drag a +1 `nC` charge (red) from the small box at the bottom and put it somewhere near the middle. Now remove it from the staging area (put it back in the storage box at the bottom), and repeat the exercise for the negative charge (blue). Describe the similarities and diffences in words. You do not have to be technical, but make your answers precise.

ii. Starting over, drag a positive charge into the central area again and leave it there. Next, add a negative charge next to it to the left. Describe the shape of the resulting field as you change the separation.

The little yellow dot labeled "Sensors" are what physicists call "test charges". A test charge is a fictitious charge that, when placed in the electric field, feels a force in the direction of the field but is assumed to be of such negible charge, that it does not contribute to the electric field. (In reality it is not possible to sense a field without affecting it, but you can limit the effect.)

iii. After playing with a test charge in the dipole field, move it so that it the force it showes is parallel to the line connecting the charges. Leave the test charge there, note the size and direction of the force it feels, and then remove the negative charge. Explain how and why the force on the test charge changes. Put the negative charge back and then remove it again repeatedly until you have convinced yourself that the size of the force it feels is much smaller with the negative charge there than without it. What does this mean for the magnitude of the electric dipole field as a function of distance r from the dipole? Your answer will be qualitatitve, but try to make it precise. For example, how would you change the statement "The magnitude of the monopole field falls off like $1/r^2$." to one appropriate for the dipole field?

1.1.14 Answer 6

In the previous exercise, we modeled an *electric* dipole field using two electric monopoles (charges): one positive and one negative as the two are brought close together. Although there are no magnetic monopoles, the magnetic dipole field has the same mathematical form as the electric one as long as the separation between monopoles is kept very small.

Now, classically, the electron is a dipole in the magnetic field of the proton or *vice versa*. (From the definition of the dipole moment, we expect the dipole moment of the proton to be much smaller than that of the electron. (Why? Hint: look at the formula for the dipole moment of a spinning point charge.) According to the formula for the energy of a dipole in a magnetic field (of the other dipole), when the spins are aligned, the energy of the system is slightly higher than when they are anti-aligned.

1.1.15 Problem 6 (cont'd)

In this problem, we will study the dipole-dipole interaction by looking a the behavior of the field resulting from two aligned dipoles and that resulting from two anti-aligned dipoles.

6b. Go back to the Phet demo and make a dipole out of two opposite charges as before.

i. Now put a second copy under it and parallel to it. Bring the two dipoles close together.

ii. In a separate window, repeat exercise 6bi but with the $\pm$ charges in the second copy switched.

iii. Make a qualitative statement about what you notice about the differnce in the two resulting fields? In what sense is one bigger than the other?

iv. Arrange 4 sensors symmetrically around your test charges in the same way in both fields. In what sense is the potential to move test charges around bigger in one field than that other.

1.1.16 Answer 6 (cont'd)

The exercise above is supposed to convince you that when the spins of the electron and proton are aligned, the energy of the system is slightly higher than when they are anti-aligned, and we want to know the wavelength of a photon emitted in a transition from the former to the latter (or, equivalently absorbed by a tranition from the latter to the former).

To recapitulate, a dipole has a magnetic field of its own. This field falls off with distance r from the dipole faster than $1/r^2$ as we saw in problem 6aiii.

1.1.17 Problem 6 (cont'd)

6c. Verify that the formula

$$\vec{B}(r) = x\frac{k}{c^2}\frac{\vec{\mu}}{r^3}$$

makes sense at least dimensionally. (*I.e.* check that the dimensions match on both sides.) Here x is just some numerical coefficient we will determine later. Do this in the following steps

- Use the Lorentz force law in `x_spectroscopy` to show that $[B] = \frac{M}{[e]T}$.
- Remind yourself of $[k]$, for example from the Coulomb Law.
- Calculate $\left[\frac{k}{c^2r^3}\right]$.
- Remind yourself of $[\mu]$ from `x_spectroscopy` or above.
- Put it all together.

1.1.18 Answer 6 (cont'd)

6c. - From the Lorentz force law in `x_spectroscopy`, $F = qvB$ allowing us to fix the dimensions of B as follows: $M \cdot L \cdot T^{-2} = [F] = [e]M \cdot T^{-1}[B]$. Solving for $[B]$ we find the result claimed. - Next, $M \cdot L \cdot T^{-2} = [F] = [k] \cdot [e]^2 \cdot L^{-2}$, so $[k] = [e]^{-2} \cdot M \cdot L^3 \cdot T^{-2}$. - Calculating $\left[\frac{k}{c^2r^3}\right] = [e]^{-2} \cdot M \cdot L^{-2}$. - From $\mu \sim \frac{e\hbar}{m}$, $[\mu] = [e] \cdot L^2 \cdot T^{-1}$. - The RHS together gives [RHS] $= [e]^{-1} \cdot M \cdot T^{-1} =$ [LHS], as claimed.

From the formula above, we get the magnetic field of the proton, and therefore the (magnetic contribution to the) energy of an electron in the proton field. Evaluating this at the Bohr radius

gives an additional contribution to the energy of

$$x\frac{k\vec{\mu}_e \cdot \vec{\mu}_p}{c^2a^3}$$

This form is pretty nice, but we can make it even more illuminating.

1.1.19 Problem 6 (cont'd)

6d. Define the energy $E_{\text{hf}} = x\frac{k\mu_e\mu_p}{c^2a^3}$ and plug in the formulæ for the magnetic moments to derive this energy in the following forms:

i. First show you can write it as

$$x\frac{ke^2}{a}\frac{1}{c^2a^2}\frac{g_p}{8}\frac{\hbar^2}{m_pm_e} = \frac{xg_p}{8}\frac{ke^2}{a}\frac{m_e}{m_p}\left(\frac{\hbar}{m_eca}\right)^2$$

ii. From `x_spectroscopy` remind yourself of the simple formulæ we derived for ionization energy of hydrogen E_1 and the find structure constant α, and use them to show that

$$E_{\text{hf}} = \frac{xg_p}{8}\frac{m_e}{m_p}|E_1|\,\alpha^2$$

iii. Recall that there were actually two configuations with energies $+|E_1|$ (aligned) and $-|E_1|$ (anti-aligned), and use it to conclude that in a transition from the former to the latter, the energy of the emitted photon must be equal to

$$\Delta E_{\text{hf}} = \frac{xg_p}{4}\frac{m_e}{m_p}|E_1|\,\alpha^2 \tag{6}$$

1.1.20 Answer 6 (cont'd)

6di. Plugging in $\mu = \frac{g_pe\hbar}{4m_p}$ and $\mu_e = \frac{e\hbar}{2m_e}$ into the formula gives

$$E_{\text{hf}} = x\frac{kg_pe^2\hbar^2}{8m_pm_ec^2a^3}$$

We can separate this as indicated in the question, but why would we? The answer is that it clarifies some things: * the factor $\frac{ke^2}{a}$ is familiar from as the potential energy of the Bohr atom in the ground state, so the rest should be dimensionless.

- The factor $\frac{g_p}{8}$ is a dimensionless number, so that leaves $\frac{1}{c^2a^2}\frac{\hbar^2}{m_pm_e}$ which can be separated democratically into
- a factor $\frac{\hbar}{m_eca}$ for the electron and
- another $\frac{\hbar}{m_pca}$ for the proton.

Are these dimensionless? You bet: the m_ec factor is a momentum and $[\hbar] = [\text{momentum}] \cdot L$, so this is dimensionless. Bells should be ringing loudly now, because we have been here before: This is the fine-structure constant α!

ii. We already covered this above: the factor $\frac{ke^2}{a} = |E_1|$. That leaves the factor $\frac{\hbar}{m_p ca}$. If we rewrite this as $\frac{\hbar}{m_p ca} = \frac{m_e}{m_p}\frac{\hbar}{m_e ca} = \frac{m_e}{m_p}\alpha$, then we get another factor of α. There is also a factor of the ratio of the masses, which is a small number $\sim \frac{1}{1800}$. Thus, we have found the requested form.

iii. The two energies for the aligned state ($\theta = 0$) and anti-aligned state ($\theta = \pi$) which differ by a factor of $\cos(0) - \cos(\pi) = 2$, so the result follows.

1.1.21 Problem 6 (cont'd)

6e. A course on classical electrodynamics suffices to fix the coefficient $x = \frac{32}{3}$.

i. Plug in the numbers to show that the energy gap is $\Delta E_{hf} = 5.88 \times 10^{-6}$ eV

ii. Convert this energy to a wavelength in centimeters. What part of the spectrum is this?

1.1.22 Answer 6 (cont'd)

6ei. We can do this various ways. We choose to define plug in the final form (6) derived in problem 6d.

[4]:
```
from astropy.constants import hbar, eps0, e, m_e, m_p
from math import pi

e = e.to(u.C)
k = 1/(4*pi*eps0)
alpha =( (k*e**2)/(hbar*c) ).decompose()
E_1 = 1/2*m_e*(alpha*c)**2

def E_hf(x: float, g: float)-> u.Quantity:
    _E = 0.25*x*g*(m_e/m_p)*E_1*alpha**2
    return _E.to(u.J)

print(f"\n|   ANSWER 6 (cont'd)\n|")

_E21 = E_hf( 32/3, 5.59 )

print(f"|   i. With x=32/3 and g_p=5.59 for the proton, as reported above, the␣
 ↪gap energy is\n|   ΔE_hf = {_E21.to(u.eV):.2e},\n|")

print(f"|   ii. The corresponding wavelength is {wavelength(_E21).to(u.cm):.0f}.
 ↪\n|   This is a {spectrum_wavelength(wavelength(_E21))}. \n")
```

```
|   ANSWER 6 (cont'd)
|
|   i. With x=32/3 and g_p=5.59 for the proton, as reported above, the gap
energy is
|   ΔE_hf = 5.88e-06 eV,
|
```

```
|   ii. The corresponding wavelength is 21 cm.
|   This is a radio wave.
```

1.1.23 Answer 6 (cont'd)

6ei. Alternatively, we can do it from the original dipole-dipole interaction.

[5]:
```
a = ( hbar/(m_e*alpha*c) ).decompose()

def mu(g,q,m: u.Quantity, s: float)-> u.Quantity:
    _mu = (g/2)*(q/m)*s*hbar
    return _mu.decompose()

mu_e = mu(2,e,m_e,1/2)
mu_p = mu(5.59,e,m_p,1/2)

def E_dipole(mu_1: u.Quantity, mu_2: u.Quantity, r: u.Quantity)-> u.Quantity:
    _E = 32/3*k/(c**2)*mu_1*mu_2/r**3
    return _E.to(u.J)

def E_hf(x: float, g: float)-> u.Quantity:
    _E = 0.25*x*g*(m_e/m_p)*E_1*alpha**2
    return _E.to(u.J)

print(f"\n|   ANSWER 6 (cont'd)\n|")

print(f"|   E_dipole = {E_dipole(mu_e, mu_p, a).to(u.eV):.2e}.\n")
```

```
|   ANSWER 6 (cont'd)
|
|   E_dipole = 5.88e-06 eV.
```

This is called the **21-cm line** or H I line. It is part of the so-called hyperfine structure of the hydrogen atom: The exceedingly closely-spaced energy levels between states in the hydrogen spectrum. The energy difference is so small that the spins can stay aligned for *millions* of years. This is an incredibly long time especially when you consider that particles in gasses typically collide *billions* of times every second. These conditions disrupt the excited state, implying that we do not see certain transitions in laboratory experiments. In typically dramatic fashion, we call these processes "**forbidden**". In the exceedingly rarified gasses encountered in space, the collision time is measured in *years*, not nanoseconds. Statistically, then, if we have enough particles some of them will have time to decay, thereby emitting a photon with a wavelength of 21-cm.

1.1.24 Problem 7

Using table 1, estimate the order of magnitude of the number of particles in an interstellar cloud.

[6]:
```
from astropy.constants import M_sun, m_p

print(f"\n|   ANSWER 7\n|")

_d = 1e17*u.m
_V = _d**3
_rho = 1e-18*u.kg/u.m**3
_m = _rho*_V

print(f"|   At a size of d ~ {_d:.0e}, the volume is of order {_V:.0e}. At a␣
↪density\n|   rho ~ {_rho:.0e}, this is a mass of {_m:.0e} or ~ {int(round(_m/
↪M_sun,-1))} solar masses.\n|   About 3/4 is protons and 1/4 is alphas (by␣
↪mass), but for this back-of-the-\n|   envelope calculation, this doesn't␣
↪matter. Dividing by the mass m_p = {m_p:.0e}\n|   of the proton, we find that␣
↪such a cloud contains on the order of N ~ {_m/m_p:.0e} or\n|   10^60 hydrogen␣
↪atoms.\n")
```

```
|   ANSWER 7
|
|   At a size of d ~ 1e+17 m, the volume is of order 1e+51 m3. At a density
|   rho ~ 1e-18 kg / m3, this is a mass of 1e+33 kg or ~ 500 solar masses.
|   About 3/4 is protons and 1/4 is alphas (by mass), but for this back-of-the-
|   envelope calculation, this doesn't matter. Dividing by the mass m_p = 2e-27
kg
|   of the proton, we find that such a cloud contains on the order of N ~ 6e+59
or
|   10^60 hydrogen atoms.
```

A statistical calculation is necessary to find the rate of emission of 21-cm radiation from an interstellar cloud, but this number of particles is so large that we can expect a measurable rate. The 21-cm line was predicted to be measurable in 1944, and was detected in 1951. It is the hallmark of ISM, and was used in 1952 to map out the distribution of matter in the galaxy. Not only can the presence of H I be detected, but the Doppler shift can be used to find radial velocity. Even the interstellar magnetic fields can be determined, because magnetic fields change the energy levels of the hydrogen atom, and therefore further shift the emission spectrum (remember the Zeeman effect? psst: the final is cumulative...).

The figure above-right shows the Pioneer plaque and the one below right shows the Voyager Golden Record. The diagrams at the top left and bottom right respectively are supposed to represent the hyperfine transition of hydrogen graphically. These objects were attached to the probes in case those probes were ever intercepted by an extraterrestrial intelligence.

1.2 Stellar formation

The birth of a star proceeds in the seven stages summarized in table 1. In the first three stages, an interstellar cloud of atoms, dust and/or molecules contracts and fragments with a fragment further

contracting into the initial stage of a protostar.

Stage	time (yr)	T_{core}(K)	T_{surface} (K)	ρ_{core} g/cm3	diameter (km)	comparable object	object
1	10^6	10	10	1.7×10^{-21}	$10^{14} \sim 3$ pc	nearest stars	interstellar cloud
2	10^4	100	10	1.7×10^{-18}	10^{12}	—	cloud fragment
3	10^5	10,000	100	1.7×10^{-12}	$10^{10} \sim$ 100 au	solar system	cloud frag/ protostar
4	10^6	1,000,000	3000	1.7×10^{-6}	$10^8 \sim 100D_\odot$	earth orbit	PMS star
5	10^7	5,000,000	4000	0.017	$10^7 \sim 10D_\odot$	—	PMS star
6	10^7	10,000,000	4500	17	$D_\odot$	sun	star
7	10^{10}	15,000,000	6000	170	$D_\odot$	sun	star (main sequence)

Table 1: Stages of stellar formation. For reference $D_\odot = 1.4 \times 10^6$ km, 1 pc $= 3.1 \times 10^{13}$ km, and the diameter of the Kuiper belt 100 AU $= 1.5 \times 10^{10}$ km. The core density of the sun is $\rho_{\text{core},\odot} = 160$ g/cm3. PMS stands for pre-main sequence.

1.2.1 Stages 1-3

We will be brief about these stages. Derivations of stated results can be found in `ss_solar_system_1,2,3`. We will start with the ISM context and progressing to smaller subregions down to that dense clump of the **giant molecular cloud** (GMC) which eventually collapses to a protostar. The structures may be represented by the following table.

	ISM complex	$\supset$	GMC	$\supset$	clump / filament	$\supset$	dense core	$\rightarrow$	protostar + disk
size	10–100 pc		20–100 pc		clump: 0.3–3 pc filament: 1–10 pc × 0.1 pc		radius 0.03–0.1 pc diameter 0.06–0.2 pc		envelope: 10^3–10^4 AU disk: 30–300 AU

	ISM complex	$\supset$	GMC	$\supset$	clump / filament	$\supset$	dense core	$\rightarrow$	protostar + disk
mass	10^3–$10^6\ M_\odot$		10^4–$10^6\ M_\odot$		10–$10^3\ M_\odot$		0.5–$10\ M_\odot$		protostar: 0.1–a few $M_\odot$ disk: 10^{-3}–$10^{-1}\ M_\odot$
number density	1–10^2 cm^{-3}		10^2–10^3 cm^{-3}		10^3–10^5 cm^{-3}		10^4–10^6 cm^{-3}		envelope: 10^5–10^8 cm^{-3} disk mid-plane can be much higher
temperature	10^2–10^4 K		10–20 K		10–20 K		10 K		envelope: 10–100 K inner disk can be much hotter
physics	large-scale ISM environment; only its coldest, densest parts are relevant for star formation		cold, self-gravitating molecular reservoir		fragmentation into local star-forming sites; often produces groups or clusters		smallest prestellar object plausibly associated with one star system or small multiple		gravity, accretion, angular momentum transport, jets/outflows, and eventual planet formation

Table 2: The structures relevant to the first few stages of star formation. The symbols should be read horizontally as "contains", "includes", "contains", and "evolves into". Thus an ISM complex contains GMCs, a GMC contains clumps and filaments, and these contain dense cores. All values are only order-of-magnitude estimates; real clouds are irregular, turbulent, magnetized, and far from spherical.

Additional notes - The **dense core** is the first entry here that is often associated with the formation of a single star system (or a small multiple system). - The last step is written with an arrow because a collapsing dense core evolves into a **protostar** plus disk. The protostar is dense shrouded core that is not yet visible from the outside (it's covered in dense material and not luminous enough to shine through).

Here is a recap (with some new tools sprinkled in) of the physics relevant to the collapse.

Astrophysics of collapse onset A GMC is self-gravitating but not in simple free fall. It is cold, molecular, turbulent, and magnetized, so gravity is opposed not only by thermal pressure but also by turbulent motions and magnetic stresses. A useful diagnostic is the virial parameter,

$$\alpha_{\rm vir} = \frac{5\sigma^2 R}{GM},$$

with $\alpha_{\rm vir} \sim 1$ indicating that self-gravity is dynamically important. In modern star-formation theory, turbulence plays a double role: on large scales it helps prevent the whole cloud from collapsing monolithically, while on smaller scales it creates overdense clumps and filaments that can become locally unstable.

Once a dense region becomes unstable, the natural collapse timescale is the **free-fall time**,

$$t_{\rm ff} = \sqrt{\frac{3\pi}{32G\rho}}.$$

Because $t_{\rm ff} \propto 1/\sqrt{\rho}$, denser substructures collapse faster than their parent cloud, which is one reason star formation proceeds hierarchically. **Fragmentation** occurs because the collapsing gas is not uniform: turbulence seeds density structure, and regions whose mass exceeds the local Jeans mass can collapse separately. While the gas remains approximately isothermal, fragmentation is comparatively easy; later heating raises the effective pressure and makes further small-scale fragmentation harder.

After a central object forms, star formation becomes largely an **accretion problem**. Gas does not fall directly onto the protostar from all directions, because even a slowly rotating core contains too much angular momentum for purely radial infall. (Recall that the angular momentum $L \propto MR^2\omega$ must be conserved, so if an initially large R decreases by a factor of 100 (stage $1 \to 2$ or $3 \to 4$) then ω must increase by a factor of $(100)^2 = 10,000$.) Instead, the infalling gas settles into a disk, and the protostar grows as mass accretes through that disk. A useful estimate for the luminosity in this phase is the accretion luminosity,

$$L_{\rm acc} \sim \frac{GM\dot{M}}{R}$$

which explains why young protostars can be luminous before sustained hydrogen fusion begins. (Here $\dot{M} = dM/dt$ is the time rate of change of the mass of the protostar.)

The central theoretical difficulty is angular momentum transport. From dense core to young star, the angular momentum must decrease by several orders of magnitude. Part of that transport happens already in the infalling envelope through primarily magnetized disk winds/jets, which remove angular momentum vertically and allow matter to spiral inward. In many modern treatments, magnetic torques and magnetized outflows are not side effects but one of the main reasons accretion can continue at all.

The temperature rise in the protostellar core and later the pre-main sequence (PMS) star/object is initially not caused by fusion, but by compressional heating. Early in collapse, the core is optically thin enough that dust thermal emission can efficiently radiate away the released gravitational energy, so the collapse is nearly isothermal. As density rises, the gas and dust become optically thick, radiative cooling can no longer keep pace with PdV heating (do you know what this means??), and the central temperature begins to increase. This produces the first **hydrostatic** core: **Hydrostatic equilibrium** is defined as when a liquid or other "plastic" (deformable) body is static (not changing shape) due to a balance between forces with one of the forces arising from pressure. In this case, the force of gravity pulling inward is balanced by a pressure "gradient" with the pressure (force per unit area) increasing as you get deeper into the center.

When the central temperature reaches roughly 2000 K, molecular hydrogen begins to dissociate; that dissociation absorbs energy, softens the equation of state (read: makes the gas/liquid easier to compress), and triggers a second collapse toward the true protostellar core.

1.3 The Kelvin-Helmholtz phase

During phase 3, the density of the material in the protostar increases to the point that it becomes opaque and radiation becomes trapped. This causes the core inner region to heat up from 100 K to 10,000 K. Meanwhile, the outer regions of the dense part of the cloud tens of au further out are still thin and transparent so the outer regions are still around the 100 K scale. A photosphere becomes sharp as the "edge" of the dense region becomes more pronounced and a **pre-main sequence** star (PMS star) has formed. The luminosity of the PMS star is a power (= energy per unit time) and that power is supplied by the gravitational energy released during contraction. This is called **Kelvin-Helmholtz** contraction.

Gravitation will continue to be the primary source of luminosity until the core reaches the 10 MK = 10^7 K needed to initiate proton-proton fusion, which is achieved only in stage 6. Thus during stages 3 through 5, the protostar is in the KH phase. Throughout this phase, the internal pressure is not yet enough to balance the gravitational attraction, so the star continues to shrink.

1.3.1 Problem 8

We can see from the HR diagram that during stage 3, the radius shrinks only moderately (the track is roughly parallel to the 100 $R_\odot$ line) and the luminosity increases only moderately (the track is roughly horizontal) as well. Let's use these two observations to estimate the time this phase takes by assuming the luminosity L and radius R stay roughly constant. Since luminosity is the rate of change of energy, we need the amount of energy available. By definition, in the Kelvin-Helmholtz phase, this energy is coming from gravitational attraction, so we will estimate it from the gravitational potential energy.

a. Use dimensional analysis to show that the gravitational potential energy of a star of radius R and mass M that formed from particles that were initially very widely separated should be

of order

$$U_g \sim \frac{GM^2}{R} \tag{7}$$

The intuition behind this is that all the particles that make up this star were separated by parsec distances, so the gravitational energy between them was very small. As they come in to a distance of radius R to make up a star, we get a lot of energy out. This energy is what heats up the star. If we do this calculation carefully, we will need calculus, but the only thing that changes is the numerical coefficient in the relation you derived from dimensional analysis. (The numerical coefficient is $\frac{3}{5}$.)

b. Calculate the gravitational potential energy for a solar mass star from your estimate.

c. The luminosity of the sun $L_\odot = 3.8 \times 10^{26}$ W. Since this is a rate of energy consumption (*i.e.* power), we can estimate the time it takes to consume any amount of energy: time ~ energy/power. Use this and the HR diagram above to estimate the order of magnitude of the amount of time a solar mass protostar will be in the Kelvin-Helmholtz contraction phase. (Compare this to table 1.)

1.3.2 Answer 8

a. From Newton's Second Law $F = ma$, the dimensions of force are $[F] = [m][a] = M \cdot L \cdot T^{-2}$. From Newton's Law of Universal Gravitation $F = Gm_1m_2/r^2$ this is also $[G] \cdot M^2 \cdot L^{-2}$, so the gravitational constant has dimensions $[G] = M^{-1} \cdot L^3 \cdot T^{-2}$. From the kinetic energy $K = \frac{1}{2}mv^2$ (or Einstein's relation $E = mc^2$), energy has dimensions $[U] = M \cdot L^2 \cdot T^{-2}$. To match the mass-dependance, we need $U \propto GM^2$. This combination has remaining dimensions $M \cdot L^3 \cdot T^{-2}$, so to cancel one power of the length, we need $U \propto GM^2/R$, as claimed.

b. See below.

c. At point 4, the luminosity $L \approx 200\ L_\odot$. Below, we plug this into the estimate for the time scale $t \sim U/L$.

[7]:
```
from astropy.constants import G, R_sun, L_sun

print(f"\n|   ANSWER 8\n|")

_U = G*M_sun**2/R_sun

_L= 200*L_sun

_t = _U/_L

print(f"|   b. With a mass of {M_sun:.1e}, and a radius of {R_sun:.1e}, we find␣
↪an energy of\n|      {_U.to(u.J):.1e}.\n|")

print(f"|   c. At a luminosity of {_L:.1e} (i.e. Joules per second), equivalent␣
↪to that of\n|      200 Suns, this translates to a time-scale of {_t.
↪decompose():.1e}. In more intuitive\n|      units, this is {_t.to(u.yr):.1e}.␣
↪According to tabel 1, this is exactly the right\n|      dex for phase 3.\n")
```

```
|   ANSWER 8
|
|   b. With a mass of 2.0e+30 kg, and a radius of 7.0e+08 m, we find an energy
of
|      3.8e+41 J.
|
|   c. At a luminosity of 7.7e+28 W (i.e. Joules per second), equivalent to that
of
|      200 Suns, this translates to a time-scale of 5.0e+12 s. In more intuitive
|      units, this is 1.6e+05 yr. According to tabel 1, this is exactly the
right
|      dex for phase 3.
```

From this calculation, we confirm that we understand the physics responsible for the contraction and luminosity of the Kelvin-Helmholtz phase.

1.3.3 Stages 4 and 5: T Tauri phase

Checking against our table, we confirm that the stage 3-to-4 handover point has $T = 3000$ K and radius 100 $R_\odot$, so the protostar is ~100 times the size of the main sequence star it will finally settle down to. At this stage, we start down the **Hayashi track**. The Hayashi limit is defined as the line across which a star goes from being fully convective as opposed to forming a radiation zone. To the left of the line, internal temperatures are so high that photoionization becomes significant and energy is transported primarily through radiation. To the right of the line, convection is so great that large quantities of thermal energy are rapidly transported to the surfact. Hayashi showed, that being even a small amount to the right of this line, the cooling effect on the interior is so great, that the star quickly adujsts back to the **Hayashi temperature** (a.k.a. the Hayashi limit): The lowest temperature that a fully convective star will be hydrodynamically stable.

Thus, in this phase of protostellar evolution, the star is fully convective (a radiation zone has not yet formed around the core) and is continuing to contract under the influence of gravity. It's surface temperature stays roughly the same, so it's luminosity drops (why? ... at this point, you should be instinctively checking statements I make against things you already understand), and the track is amost vertical.

The name of this pre-main sequence (PMS) object is a **T Tauri star** in honor of the first PMS object observed in the constellation Taurus (shown at left). It is a **variable** star meaning that its luminosity is changing in time. Thick convection zones transport energy to the surface which often results in violent eruptions. During this phase, the protostar gives rise to the strong protostellar winds responsible for clearing the protostellar system of interplanetary gas. Early in this phase, the winds are collimated into **jets** resulting in **bipolar outflow** from the poles and perpendicular to the protoplanetary disk. Often, these streams of ejected material collide with the protostellar medium around it making large and visible shock fronts called **Herbig-Haro objects**. Herbig-Haro object HH30 is shown in the `gif` to the right.

Here is a link to a recent simulation by Starforge demonstrating star formation from an interstellar cloud. (See also this.) In the simulation we see many instances of mass outflows of the T Tauri stars.

1.3.4 Problem 9.

Watch the Starforge simulation linked above. When do the first protostars enter the T Tauri phase? At some point near the end, there is a large flash. When does this flash happen and what do you think is happening then?

1.3.5 Answer 9.

The first T Tauri-phase stars occur around 01:00 and the supernova is around 03:00.

1.4 Stage 6

As the T Tauri star progresses down the Hayashi track with a roughly-constant surface temperature the luminosity is decreasing in accordance with the Stefan-Boltzmann law for a shrinking object. The heat generated by this contraction acts to raise the core temperature. This increase, in turn, causes the opacity of the core to drop below a point where radiative energy transportation becomes competitive: a radiation zone starts to form and then grows in size. This transfers us off the Hayashi track onto the **Henyey track**.

At this point, we have a radiative zone inside of a convective outer region. Although the Kelvin-Helmholtz contraction is still ongoing, core temperature and pressure have rising to the point that hydrogen fusion begins at 10 `MK` = 10,000,000 `K`. As contraction proceeds the luminosity begins to transition between gravitationally-powered to fusion-powered.

Although the protostar has become a star, the internal pressure is still not balancing the gravitational attraction, and contraction continues until stage 7, where the star finally reaches **hydrostatic equilibrium**.

1.5 Stage 7

At the end of stage 6, the protostar has become a star, but the internal pressure is still not balancing the gravitational attraction, and contraction continues until stage 7, where the star finally reaches **hydrostatic equilibrium**. This is the final condition needed to define a **zero-age main sequence** star.

What defines a **star on the main sequence** is that * it "burns" hydrogen specifically (*i.e.* fuses it into helium), and
* it is in hydrostatic equilibrium.

Eventually all such stars must burn through their hydrogen. The rate at which they do so is proportional to their luminosity (see problem 8).

1.5.1 Problem 10

Estimate the time a solar-mass star with spectral type G will remain on the main sequence. The fusion of protons in the sun generates $\Delta E_{pp} = 1 \times 10^{-12}$ `J` per proton.

a. Calculate the number of protons in the Sun by assuming 75% of it is made of protons.

b. Assume 20% of this is available for fusion at a rate sustaining the solar luminosity $L_\odot$ of the Sun, and use this to estimate the time this star will remain on the main sequence.

[8]:
```
print(f"\n|   ANSWER 10\n|")

_M = 3/4*M_sun

_N = _M/m_p

print(f"|   a. 3/4 of the mass of the Sun is about {_M:.1e} giving {_N:.1e}␣
↪protons.\n|")

_Neff = .2*_N

_E = _N*1e-12*u.J

_t = _E/L_sun

print(f"|   b. 20% of these protons is about {_Neff:.1e} giving and energy of␣
↪{_E:.1e}.\n|      At a rate of {L_sun:.1e}, this is a lifetime {_t:.1e} or {_t.
↪to(u.Gyr):.1e}. \n|")
```

```
|   ANSWER 10
|
|   a. 3/4 of the mass of the Sun is about 1.5e+30 kg giving 8.9e+56 protons.
|
|   b. 20% of these protons is about 1.8e+56 giving and energy of 8.9e+44 J.
|      At a rate of 3.8e+26 W, this is a lifetime 2.3e+18 J / W or 7.4e+01 Gyr.
|
```

1.5.2 Mass-Luminosity/Radius relations

1.5.3 Problem 11

Consider the diagram at right (which I have borrowed from Carroll and Ostlie).

a. Describe what you see.

b. Use it to construct a formula for L as a function of M.

1.5.4 Answer 11

a. It appears to be a diagram of the luminosity of a collection of stars plotted against the mass of those stars.

- From the legend, I gather that these stars are in various types of binary systems, which is presumably how they determined the masses.
- I also notice that all the stars are very different in mass ranging from 0.1 $M_\odot$ to $10^{1.25}$ $M_\odot \approx$ 18 $M_\odot$. This is confirmed by "OB systems" label in the legend as OB stars are very hot an luminous. Even across such large mass ranges, the relation is quite linear.

b. Taking the run from -0.5 to 1.25 and the rise from -2 to 4.5, I'm getting a slope of $\alpha = 6.5/1.75 \approx 3.7$. The relation is, therefore $L/L_\odot = (M/M_\odot)^\alpha$ with this value for α.

This is called the **mass-luminosity relation**. Actually, there are multiple such relations that are all of the form

$$L \propto M^{\alpha} \tag{8}$$

for some constant α. The value of α quoted in the book is $\alpha \approx 4$. I usually see this quoted to be $\alpha \approx 3.5$. Later in the course when we look at the particulars of stellar evolution, we may come back and tweak this later. I will stick with $\alpha \approx 3.7$, but let me briefly sketch the reason such a relation is not entirely unexpected.

We will use three main ideas that may or may not be good approximations/models: * Hydrostatic equilibrium: pressure gradients are what are holding up the star against gravitational collabpse. This is reasonable because it is one of the two defining conditions for a main-sequence star. * Ideal gas equation of state. This is a bit iffy, but if we can think of a star as a giant ball of ideal gas, then $PV = Nk_BT$. * Radiative diffusion: The energy transport out of the core is carried by radiation. This is very iffy for very massive or very low-mass stars, as we will discuss later.

The first assumption lets us say that the pressure $P = \text{energy}/V$ at the core is balancing $U_g = GM^2/R$, thus

$$P \sim \frac{GM^2}{R^4}$$

From the second assumption, this is equal to $P = \frac{\rho}{m_p} k_B T$ where ρ is the mass density, so $\frac{\rho}{m_p}$ is the number density. This gives the core temperature as

$$k_B T_c \sim \frac{G m_p M}{R}$$

The final assumption allow us to say that the energy transport out of the core has a flux given by the Stefan-Boltzmann law $F = \sigma_B T^4$. Unfortunately, we cannot blindly plug in SB in the form $L \sim R^2 T^4$ because this formula was appropriate for the surface of the star and using it would only be correct if $T_{eff} = T_c$ which is very much wrong, as we can see from table 1. To proceed this way, we need to talk about *opacity* which is how not-transparent the material is to radiation. What lowers the temperature as we go outward is precisely the opacity κ that we are missing. It is usually a reasonable assumption to take the opacity κ to be constant, but this factor has dimensions $[\kappa] = L^2/M$, so its existence will change the dimensional analysis, because the combination $\frac{\kappa M}{R^2} = \kappa \rho R$ is dimensionless. Specifically, the flux is modified by opacity to

$$F \sim \frac{\sigma_B T_c^4}{\kappa \rho R}$$

where we are taking κ and ρ to be constant. Note that while this is a reasonable guess (*e.g.* as opacity increases, the flux decreases), a derivation of this result is needed in order to justify this formula. (It is not a difficult derivation, but this packet is already too long.)

In any case, if we accept this modification, we can proceed as before, as the luminosity now follows

$$L \sim R^2 F \sim \frac{\sigma_B R T_c^4}{\kappa \rho} \sim \frac{\sigma_B R^4}{\kappa M} \left(\frac{G m_p M}{R} \right)^4 \propto \frac{M^3}{\kappa}$$

Notice that it is very convenient that the radius has canceled since we do not (yet) know the relation (if any) between mass and radius.

As we already discussed in Problem 10, the luminosity represents the amount of energy emitted per unit time (the rate of burning), so the lifetime t of a star is inversely proportional to the luminosity

$t \propto 1/L$. Since there is more fuel to burn when there is more mass, it is also proportional to the mass $t \propto M$. Taken together with the mass-luminosity relation (8), we get that the lifetime of a star is inversely proportional to the cube of the mass of the star:

$$t \propto \frac{1}{M^{\alpha-1}} \tag{8}$$

This means that the more massive the star, the shorter it stays on the main sequence branch.

1.5.5 Problem 12

An oft-cited estimate for the total lifetime of the sun is $t = 1 \times 10^{10}$ `yr` 10 `Gyr`. Use this to estimate the lifetime of a 15-solar-mass main sequence star. What about a $60M_\odot$ star?

[9]:
```
print(f"\n|   ANSWER 12\n|")
t_sun = 10_000_000_000
t_15 = 15**(-3)
t_60 = 60**(-3)
print(f"|   A 15-solar mass star has a lifetime that is {round(1/t_15)} times␣
↪shorter than our Sun, \n|   or {int(round(t_15*t_sun/1e6))} million years.\n|")
print(f"|   A 60-solar mass star has a lifetime that is {round(1/t_60)} times␣
↪shorter than our Sun, \n|   or {int(round(t_60*t_sun/1e3,-1))} thousand years.
↪\n")
```

```
|   ANSWER 12
|
|   A 15-solar mass star has a lifetime that is 3375 times shorter than our Sun,
|   or 3 million years.
|
|   A 60-solar mass star has a lifetime that is 216000 times shorter than our
Sun,
|   or 50 thousand years.
```

1.5.6 Problem 13

Go back to table 1 and add up all the time it takes to form a star, that is the time from the beginning of stage 1 until the end of stage 6. Compare this to the time the star stays on the main sequence. What percentage of the latter time is the former?

[10]:
```
print(f"\n|   ANSWER 13\n|")
print(f"|   After the main sequence at 10 Gyr, stages 5 and 6 are the longest at␣
↪10 Myr;\n|   each is smaller by a factor of 1000. As a fraction, this means␣
↪that a star is on\n|   the main sequence 1/(1+2/1000) of the time, or␣
↪percentage-wise\n|   {100/(1+2/1000):.1f}% of the time.\n")
```

```
|   ANSWER 13
|
|   After the main sequence at 10 Gyr, stages 5 and 6 are the longest at 10 Myr;
```

```
|   each is smaller by a factor of 1000. As a fraction, this means that a star
is on
|   the main sequence 1/(1+2/1000) of the time, or percentage-wise
|   99.8% of the time.
```